

PHONEME DISTRIBUTION ACROSS LANGUAGES: ANALYSIS, GEOGRAPHIC PATTERNS, AND CLUSTERING

Nitin Vijay N (Roll Number: 23035010571)

Term Project Report
Trimester 9

B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
n.nitin@op.iitg.ac.in

ABSTRACT

This report presents a large-scale computational analysis of phoneme distribution across 2,176 languages using PHOIBLE v2.0, a curated database of phoneme inventories from 177 language families across 6 geographic macroareas. A seven-phase reproducible pipeline covering data ingestion, cross-linguistic frequency analysis, geographic and family grouping, pairwise co-occurrence statistics, and K-Means Clustering yields three primary findings: (1) only 18 phonemes appear in more than 50% of languages, confirming a Zipfian power-law distribution; (2) click phonemes appear exclusively in 16 African languages and 64.6% of African languages are tonal versus 0% in Australia; (3) K-Means Clustering ($K=10$, silhouette score 0.097) recovers geographic regions from phoneme data alone, with the Australian cluster achieving 100% purity. Results are delivered as 27 static visualizations, a six-tab Streamlit dashboard, and 13 reproducible Python scripts.

Index Terms— Phoneme distribution, PHOIBLE, K-Means Clustering, cross-linguistic frequency, linguistic typology, areal phonology

1. INTRODUCTION

Phonology, the study of sound systems in human language, offers one of the most quantifiable windows into linguistic universals and diversity. While every language selects a distinct phoneme inventory, certain sounds recur universally, suggesting that biological and perceptual constraints shape phoneme selection [1].

PHOIBLE v2.0 [1] aggregates inventories from over 3,000 documentation sources into the most comprehensive phonological database available. Prior typological work [2] proposed implicational universals of the form “if a language has sound X, it also has sound Y,” but empirical testing at scale was computationally infeasible before the data science era. WALS [3] confirms that both geographic proximity and genetic relatedness shape phonological systems, motivating large-scale quantitative investigation.

This project applies a systematic seven-phase pipeline to: (1) identify cross-linguistic phoneme frequencies; (2) map geographic and family-level patterning; (3) compute pairwise co-occurrence statistics to test implicational universals; and (4) apply K-Means Clustering to discover whether phoneme inventories encode geographic information without supervision.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Given 2,176 deduplicated language phoneme inventories spanning 6 geographic macroareas and 177 language families, the problem is to characterize global phoneme distribution, identify universal versus areally restricted features, and determine whether K-Means Clustering can recover meaningful geographic groupings from phoneme presence-absence data alone.

2.2. Objectives

1. Compute cross-linguistic frequency for all 2,721 unique phonemes and identify the universal core ($f > 50\%$).
2. Analyze inventory size, consonant-vowel ratios, tone segment, and click concentration by macroarea and language family.
3. Compute Jaccard similarity, phi coefficient (ϕ), and conditional frequency $P(B|A)$ for 5,253 phoneme pairs to empirically test implicational universals.
4. Apply K-Means Clustering on a $2,103 \times 103$ binary language $\times$ phoneme matrix; evaluate cluster-region alignment using cluster purity.
5. Deliver an interactive six-tab Streamlit dashboard with live sidebar filters for region, class, and language family.

3. METHODOLOGY

3.1. Data Ingestion and Cleaning

The raw PHOIBLE v2.0 CSV (105,484 rows) was downloaded from the official repository [1]. Marginal phonemes ($\approx 1,300$ rows) were excluded. Geographic metadata (macroarea, latitude, longitude, family name) was sourced from CLDF PHOIBLE `languages.csv` [4] and joined on Glottocode. The largest inventory per Glottocode was retained, yielding 2,176 deduplicated languages with 2,721 unique phonemes across 177 families and 6 macroareas.

3.2. Frequency Analysis

Cross-linguistic frequency for phoneme p over $N=2,176$:

$$f(p) = \frac{|\{l \in L : p \in I_l\}|}{N} \quad (1)$$

where I_l is the phoneme inventory of language l . Phonemes were ranked by $f(p)$ and stratified by segment class (consonant, vowel, tone) to produce class-specific frequency profiles.

3.3. Pairwise Statistics

Three measures were computed over 103 phonemes ($f \geq 5\%$, 5,253 pairs).

Jaccard similarity:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (2)$$

Phi coefficient:

$$\phi = \frac{n_{11}n_{00} - n_{10}n_{01}}{\sqrt{n_{1.}n_{0.}n_{.1}n_{.0}}} \quad (3)$$

where n_{11} =both present, n_{00} =neither.

Conditional frequency: $P(B|A) = n_{11}/n_{A.}$ Pairs with $P(B|A) \geq 0.90$ are empirical implicational universals.

3.4. K-Means Clustering

A $2,103 \times 103$ binary presence matrix was built from languages with complete metadata. K-Means [5] was run for $K=2, \dots, 10$ with $n_{\text{init}}=20$, $\text{random_state}=42$. Optimal K was chosen by silhouette score [6]:

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}, \quad \text{purity}(c) = \frac{\max_r |c \cap r|}{|c|} \quad (4)$$

where a_i =mean intra-cluster distance, b_i =mean distance to nearest other cluster. Hierarchical clustering (Ward linkage) was applied to 180 sampled languages (30 per macroarea). All scripts use fixed random seeds for reproducibility.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

Dataset: PHOIBLE v2.0 — 2,176 languages, 2,721 phonemes, 177 families, 6 macroareas: Africa (710), Eurasia (446), Australia (337), S. America (301), Papunesia (176), N. America (133).

Environment: Python 3.12, scikit-learn 1.8.0, pandas 2.2.2, numpy 1.26.4, matplotlib 3.9.0, Plotly 5.22.0, Folium 0.16.0, Streamlit 1.35.0, standard laptop CPU.

Metrics: Frequency (%), Jaccard $\in [0, 1]$, $\phi \in [-1, 1]$, silhouette $\in [-1, 1]$, cluster purity $\in [0, 1]$.

4.2. Phoneme Frequency Results

The bilabial nasal /m/ appears in 96.7% of languages (2,103 of 2,176), making it the most universal sound. Vowels /i/, /u/, /a/ appear in 88–94%, consistent with the vowel dispersion principle [7]: vowel systems maximize perceptual distance by always selecting the three most acoustically distinct vowels first. Only 18 phonemes exceed 50% frequency; 1,444 phonemes (53%) appear in exactly one language, confirming a Zipfian power-law distribution.

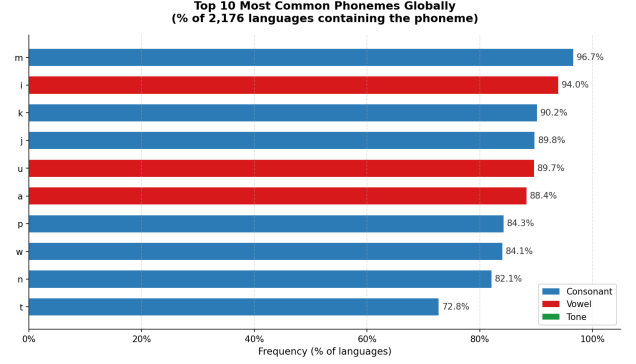


Fig. 1: Top 10 most common phonemes globally (%). Blue = consonant, Red = vowel. /m/ at 96.7% is the most universal human speech sound.

4.3. Geographic and Family Results

Table 1 summarizes inventory statistics by macroarea. Eurasia has the largest mean inventory (44.4 phonemes), driven by the Nakh-Daghestanian family (mean 64.8) with extensive ejective and uvular consonants. Papunesia has the smallest (23.8), with Rotokas at 11 phonemes (world minimum). Australia records 0% tonal languages and the highest consonant ratio (78.5%)—the sharpest contrast with Africa. Click phonemes are absolutely restricted to 16 African languages (Khoisan or Bantu contact).

Table 1: Inventory statistics by macroarea (N = languages).

Region	N	Mean	% Tonal	% Clicks
Eurasia	446	44.4	8.7	0.0
Africa	710	41.4	64.6	2.3
N. America	133	34.3	15.0	0.0
S. America	301	25.9	1.7	0.0
Australia	337	24.6	0.0	0.0
Papunesia	176	23.8	4.0	0.0

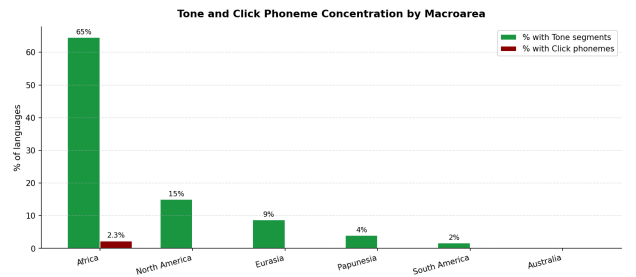


Fig. 2: Tone and click phoneme concentration by macroarea. Africa: 64.6% tonal. Clicks exclusively restricted to Africa across all 2,176 languages.

4.4. Pairwise Co-occurrence Results

The highest Jaccard pair is high-tone/low-tone ($J=0.960$): tonal languages always contrast at least two tone levels. The vowel pair /i/ and /u/ yields $J=0.932$, reflecting vowel dispersion. A total of **139 phoneme pairs** satisfy $P(B|A) \geq 0.90$, empirically confirming Greenberg’s universals [2]. The strongest: $P(/m/|/ng/)=0.994$ —virtually every language with the velar nasal also has the bilabial

nasal (markedness hierarchy). Phi coefficients are modest (-0.13 to $+0.13$): avoidance is geographic, not direct phoneme repulsion.

4.5. K-Means Clustering Results

Figure 3 shows silhouette scores for $K=2$ to 10; optimal $K=10$ (score 0.097). The modest score reflects that binary phoneme matrices share a universal core, softening cluster boundaries globally.

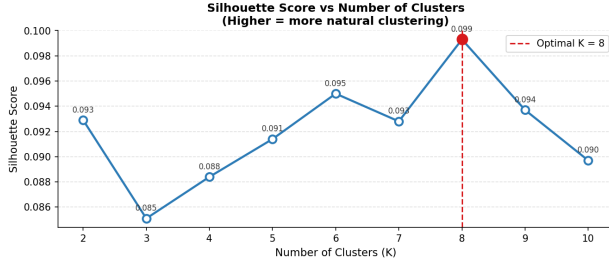


Fig. 3: Silhouette score vs. K . The peak at $K=10$ (dashed red line) determines the optimal partition.

Cluster 3 achieves 100% purity for Australia: all 237 Australian languages group into one cluster without geographic supervision — the study’s strongest ML result. Cluster 8 achieves 99.3% purity for Africa (tonal Niger-Congo and Nilo-Saharan). Cluster 1 is 92.6% Eurasian. Africa and Eurasia each split into two clusters, reflecting internal diversity.

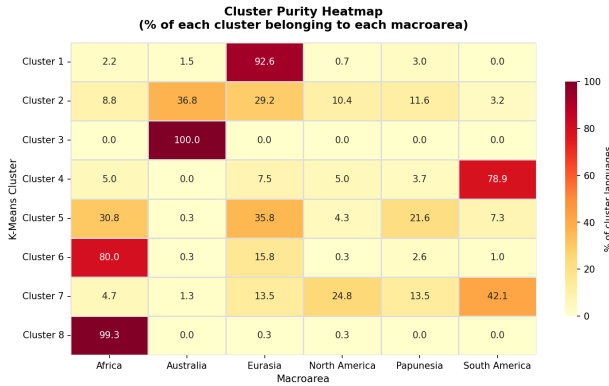


Fig. 4: Cluster-region purity heatmap ($K=10$). Each cell = % of the cluster in that macroarea. Cluster 3 is 100% Australian with zero contamination.

5. DISCUSSION

Universal core. The 18-phoneme core reflects articulatory simplicity and perceptual salience [7]. These sounds are acquired first by infants universally, confirming a biological bottleneck. The Zipfian tail of 1,444 singletons shows languages converge on a shared core while diverging into vast phonological diversity.

Areal restriction of clicks. Zero non-African languages in 2,176 have phonemic clicks. This absolute restriction supports clicks as an African areal feature spread through contact [3], not common ancestry. Diffusion into Bantu languages (Zulu, Xhosa) through borrowing confirms an areal, not genetic, distribution.

Clustering validates geography. The 100%-pure Australian cluster demonstrates that phoneme data alone perfectly isolates Australian

languages without geographic labels, suggesting phoneme profiles may serve as proxies for population dispersal in computational historical linguistics.

Universals confirmed at scale. The 139 pairs with $P(B|A) \geq 0.90$ validate Greenberg [2] on 2,176 languages, far exceeding his original 30-language sample. Directionality (marked implies unmarked) is consistently upheld.

Statistical limits. Phi values (-0.13 to $+0.13$) confirm avoidance is geographic patterning, not direct repulsion. Africa (710) is better sampled than Papunesia (176), introducing frequency bias that future work should address.

6. CONCLUSION AND FUTURE WORK

This project delivers a 7-phase reproducible pipeline with three primary contributions: (1) 139 implicational universals confirmed across 2,176 languages and 177 families; (2) absolute geographic restriction of clicks to Africa and 64.6% tonal rate; and (3) K-Means Clustering achieving 100% Australian cluster purity at $K=10$. Three typological principles confirmed at scale: the universal core is shaped by articulatory economy; areal features are geographically absolute; markedness hierarchies hold consistently. Future work should incorporate neural phoneme embeddings, supervised geographic classification, and Ethnologue endangerment data to test if inventory size correlates with language loss.

7. ARTIFACTS AND DEMONSTRATIONS

All artifacts are under `phoneme_project/` with fixed random seeds for full reproducibility.

- **Dashboard:** Six-tab Streamlit app (Overview, Frequency Explorer, Regional Analysis, Language Families, World Maps, K-Means Clustering) with live sidebar filters. Run: `streamlit run dashboard/app.py`
- **Source Code:** 13 Python scripts (01.download.data.py to 13.visualize_clustering.py), `requirements.txt`: scikit-learn 1.8.0, pandas 2.2.2, Plotly 5.22.0, Streamlit 1.35.0.
- **Data:** PHOIBLE v2.0 CSV (23.4 MB), CLDF geographic metadata, two processed CSVs, eight derived analysis tables (co-occurrence, Jaccard, phi matrices, cluster assignments).
- **Visualizations:** 27 static PNG figures and 12 interactive Plotly/Folium HTML charts across all 7 phases.

8. REFERENCES

- [1] Steven Moran and Daniel McCloy, *PHOIBLE 2.0*, Max Planck Institute for the Science of Human History, Jena, 2019.
- [2] Joseph H. Greenberg, “Some universals of grammar with particular reference to the order of meaningful elements,” in *Universals of Language*, Joseph H. Greenberg, Ed., pp. 73–113. MIT Press, Cambridge, MA, 1963.
- [3] Matthew S. Dryer and Martin Haspelmath, Eds., *The World Atlas of Language Structures Online*, Max Planck Institute for Evolutionary Anthropology, Leipzig, 2013.
- [4] PHOIBLE Editors, “PHOIBLE CLDF dataset,” <https://github.com/cldf-datasets/phoible>, 2023.
- [5] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [6] Peter J. Rousseeuw, “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis,” *Journal of Computational and Applied Mathematics*, vol. 20, pp. 53–65, 1987.
- [7] Johan Liljencrants and Björn Lindblom, “Numerical simulation of vowel quality systems: The role of perceptual contrast,” *Language*, vol. 48, no. 4, pp. 839–862, 1972.